

Literature-implied answer: Banach-space subcase of Markov-to-fork convexity

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Status. Literature-implied answer (Banach-space subcase). This note records a scoped consequence of known literature for Problem `pb:markov->fork` in Baudier–Gartland, *Umbel convexity and the geometry of trees*, arXiv:2103.16011. It does not solve the problem for arbitrary metric spaces.

Source question. In Section 6, Problem `pb:markov->fork`, Baudier–Gartland ask whether Markov p -convexity implies fork p -convexity. Immediately before the problem they explain that Markov p -convexity averages over all tree levels, whereas fork p -convexity involves dyadic-level terms.

Supporting theorem. Mendel–Naor, *Markov convexity and local rigidity of distorted metrics*, arXiv:0803.1697, Theorem 1.3, prove that a Banach space is Markov p -convex if and only if it admits an equivalent norm whose modulus of uniform convexity has power type p . Baudier–Gartland cite this result as Theorem `thm:MN-LNP`.

Identification. Let X be a Banach space and let $p \geq 2$. If X , with its original norm metric, is Markov p -convex, then by Mendel–Naor it has an equivalent p -uniformly convex norm. Baudier–Gartland recall the p -fork point inequality in their equation `eq:qfork`, and in Section 5 note, citing Mendel–Naor Lemma 2.3, that a p -uniformly convex Banach norm satisfies the p -fork inequality. The full p -fork point inequality implies the relaxed fork inequality used in their Theorem `thm:fork`, hence the equivalent norm metric is fork p -convex. Fork p -convexity is a bi-Lipschitz invariant, so passing back to the original equivalent Banach norm preserves the conclusion.

Conclusion. For Banach spaces in the renorming range $p \geq 2$, Markov p -convexity does imply fork p -convexity. The general metric-space question in Baudier–Gartland remains open and unclaimed here. The supporting theorem predates the source problem; this packet records an agent-identified subcase, not an explicit later answer by the supporting authors.

References.

- F. P. Baudier and C. Gartland, *Umbel convexity and the geometry of trees*, arXiv:2103.16011.
- M. Mendel and A. Naor, *Markov convexity and local rigidity of distorted metrics*, arXiv:0803.1697; J. Eur. Math. Soc. 15 (2013), 287–337.